FORM TP 2015045

TEST CODE **01207020**

MAY/JUNE 2015

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, answers should be illustrated with diagrams.
5. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
6. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions in this section.

Write your answers in the spaces provided in this booklet.

1. Students of an ecology class conduct a field study in an area located on the boundary between a savannah and a forest. In addition to using various sampling methods to estimate species abundance, they observe the feeding patterns of the organisms.

(a) (i) Describe TWO sampling methods which the students could use to estimate the abundance of **plant species** in the area.

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(4 marks)

(ii) Describe ONE method they could use to collect data on the mobile **animals** which inhabit the area.

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(2 marks)

(b) The students record their data in a table similar to Table 1.

TABLE 1: DATA ON ORGANISMS IN THE FIELD STUDY

Species Names	Estimated Numbers in 100 m^2	Feeding Behaviour
Savannah grass	1000	
Grasshopper	50	Eats grass and leaves
Caterpillar	25	Eats leaves
Poui tree	2	
Hawk	3	Eats woodpeckers and lizards
Woodpecker	5	Eats caterpillars
Lizard	15	Eats caterpillars

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Use the data in Table 1 to construct EACH of the following:

(i) A food web

(4 marks)

(ii) A pyramid of numbers to show the feeding relationships among the organisms living in the area

(2 marks)

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(c) The students also collect samples of topsoil from an area under some trees.

(i) Describe an experiment that the students could use to investigate the water-holding capacity of the soil.

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(3 marks)

(ii) State FOUR ways in which soil is important to living organisms.

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(4 marks)

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(d) Figure 1 shows the apparatus that the students use to extract small animals (invertebrates) living in the soil.

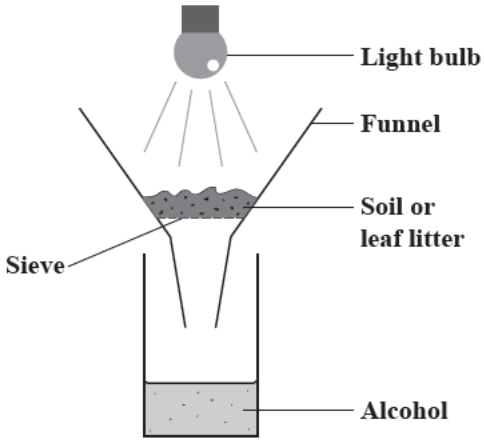

Figure 1. Apparatus for extracting small animals

Outline how EACH of the following functions in the apparatus:

Light bulb .....

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(2 marks)

Alcohol .....

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(1 mark)

(e) Fungi and bacteria are found in forest soils. Explain the role of these organisms in the cycling of nutrients in the forest.

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(3 marks)

Total 25 marks

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2. (a) The following chemical equation for photosynthesis represents the reactants used and the products formed.

$$6CO_2 + 6H_2O \longrightarrow C_6H_{12}O_6 + 6O_2$$

(i) Name TWO conditions, which are NOT shown in the equation, that are essential for this process.

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(2 marks)

(ii) Name the gas given off in this process and state which of the reactants produce this gas.

Gas .....

Reactant .....

(2 marks)

(iii) Explain TWO ways in which plants utilize the glucose produced during this process.

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(4 marks)

(iv) State THREE environmental factors which affect photosynthesis.

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(3 marks)

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(b) Unlike plants, animals are unable to synthesize glucose. A boy eats a meal of cassava. Explain how his body converts the starch in the cassava to glucose. Name the organs and the enzymes involved in this process.

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(4 marks)

Total 15 marks

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3. Figure 2 is a diagram of the human female reproductive system.

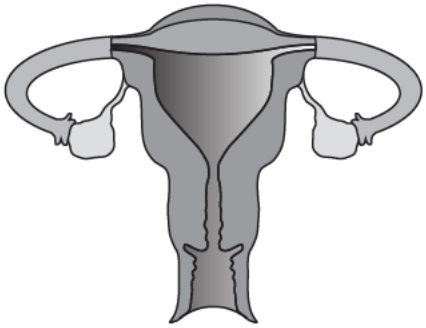

Figure 2. Structure of the human female reproductive system

(a) Label the organs in Figure 2 to show where

(i) the female gametes (ova) are made **(1 mark)**

(ii) fertilization would normally occur **(1 mark)**

(iii) implantation of a fertilized gamete would normally occur. **(1 mark)**

(b) Outline the mechanism by which male and female gametes come together in the human reproductive system to form a zygote.

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(3 marks)

(c) (i) Name the type of cell division by which female gametes are made.

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(1 mark)

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(ii) Describe TWO ways in which the type of cell division named in (c) (i) differs from that by which the embryo grows and develops after fertilization of an ovum.

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(4 marks)

(d) Embryonic stem cells are undifferentiated cells formed as the embryo grows and develops. These cells are capable of differentiating into specialized cells of tissues and organs in the human body.

(i) Suggest TWO reasons why cell specialization is important.

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(2 marks)

(ii) Explain why embryonic stem cells can be used in the treatment of physiological diseases.

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(2 marks)

Total 15 marks

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SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided at the end of each question.

4. Albinism is seen in persons who are homozygous for the recessive allele of a certain gene. This gene codes for the production of the skin pigment, melanin. Persons who inherit the dominant allele of this gene produce the normal amount of melanin for their race.

(a) Distinguish between the following paired terms: allele/gene; dominant/recessive; homozygous/heterozygous. **(6 marks)**

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(b) (i) Use a genetic diagram to show how a couple with normal pigmentation may produce an albino child. Use the following symbols to represent the alleles: A – normal; a – albino. **(5 marks)**

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(ii) Suggest TWO precautions that albino persons living in the Caribbean should take when going outdoors. **(2 marks)**

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(c) Observation of the members of a population shows that there is variation in skin colour. Suggest THREE ways by which this type of variation could occur. **(2 marks)**

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Total 15 marks

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5. The central nervous system coordinates the link between stimuli and responses.

(a) Make a labelled diagram of a horizontal section through the human eye and describe how the eye enables us to see. **(6 marks)**

Space for diagram

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(b) Discuss the effects of alcohol abuse on the human body's ability to respond to stimuli and maintain homeostasis. **(9 marks)**

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Total 15 marks

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6. (a) Name THREE characteristics which distinguish a biological species and THREE factors that lead to the formation of a new species. **(6 marks)**

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(b) The distribution of variant forms of the peppered moth, *Biston betularia*, changed between pre-industrial and post-industrial Britain. Account for the increase in the numbers of the dark form in the post-industrial era. **(3 marks)**

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(c) Use the theory of natural selection to explain antibiotic resistance. **(6 marks)**

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Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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